I^{2}SB through the ImMAP lens:

the Brownian bridge as a ladder of conditional denoising problems, and Tweedie ascent toward the MAP estimate given $\boldsymbol{x}_1$

ImMAP project notes

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Abstract

I^{2}SB trains a network on a Brownian bridge between paired endpoints $(\boldsymbol{x}_0, \boldsymbol{x}_1)$ and samples with a DDPM-style reverse posterior. We give a different reading of the same objects. With the starting point $\boldsymbol{x}_1$ known, every bridge state is, after an affine change of coordinates, a Gaussian *denoising* problem for $\boldsymbol{x}_0$ at a noise level fixed by the bridge time. The trained regressor is therefore a conditional MMSE denoiser, $D_\theta(\boldsymbol{x}_t, t) \approx \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1]$, and by Tweedie's formula its residual is a scaled score of the noise-smoothed conditional prior $p_\sigma(\cdot \mid \boldsymbol{x}_1)$. This turns the bridge into the setting of Kadkhodaie–Simoncelli and ImMAP: an annealed, Tweedie-driven ascent toward the maximum a posteriori estimate of $\boldsymbol{x}_0$ given $\boldsymbol{x}_1$ (and, with a data term, given the measurement $\boldsymbol{y}$ too). We write that ascent directly in bridge coordinates, with noise injection that keeps every iterate on the bridge marginals, and show that I^{2}SB's own sampler is the special case with a prescribed step size and noise level. The formulas match `sb/immap_sb_ascent.py` line for line; Table 1 gives the correspondence.

Notation. Images are vectors in $\mathbb{C}^N$. Noise is circularly-symmetric complex Gaussian, $\boldsymbol{\varepsilon} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{I})$, i.e. $\mathbb{E}|\varepsilon_n|^2 = 1$ per entry (variance $1/2$ per real component); $\mathcal{CN}(\boldsymbol{\mu}, s^2\boldsymbol{I})$ has density $(\pi s^2)^{-N} \exp(-\|\boldsymbol{x}-\boldsymbol{\mu}\|^2/s^2)$. For a real function f of a complex vector we use the Wirtinger gradient $\nabla_{\boldsymbol{u}} f \coloneqq \partial f / \partial \bar{\boldsymbol{u}}$, which is half the gradient in real coordinates, $\frac{1}{2}(\partial_{\mathrm{Re}} + i\,\partial_{\mathrm{Im}})$; with it, all formulas below read exactly as in the real case. $\boldsymbol{x}_0$ is the target (the fully-sampled image), $\boldsymbol{x}_1$ the bridge's prior endpoint (in MRI: the zero-filled image $\boldsymbol{E}^H\boldsymbol{y}$, the ACS low-pass image, or a guide contrast), and $t \in [0, 1]$ the bridge time, $t = 0$ at $\boldsymbol{x}_0$.

1 The Brownian bridge

1.1 Marginals

Let $X_t = \boldsymbol{x}_0 + \sqrt{V}\, B_t$ with B_t a standard complex Brownian motion, so $\mathrm{Cov}(X_t, X_s) = V \min(t, s)\, \boldsymbol{I}$. Conditioning on the endpoint $X_1 = \boldsymbol{x}_1$ pins the path at both ends. Because (X_t, X_1) is jointly Gaussian given $\boldsymbol{x}_0$, with $\mathrm{Cov}(X_t, X_1) = Vt\,\boldsymbol{I}$ and $\mathrm{Var}\, X_1 = V\boldsymbol{I}$, the conditional law is

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_0, \boldsymbol{x}_1) = \mathcal{CN}\big((1-t)\,\boldsymbol{x}_0 + t\,\boldsymbol{x}_1,\ \sigma_{\mathrm{sb}}(t)^2\,\boldsymbol{I}\big), \qquad \sigma_{\mathrm{sb}}(t)^2 = V\,t(1-t). \tag{1}$$

Equivalently,

$$\boldsymbol{x}_t = (1-t)\,\boldsymbol{x}_0 + t\,\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t)\,\boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{I}), \quad \boldsymbol{\varepsilon} \perp (\boldsymbol{x}_0, \boldsymbol{x}_1). \tag{2}$$

The total variance V is set by the bridge *temperature* τ, $V = 4\tau^2$, so that σ_{sb} peaks at τ at the midpoint $t = \frac{1}{2}$. In the code, σ_{sb} is `std_sb` and $\sqrt{Vt}$ is the forward std `std_fwd`, with $\texttt{std_fwd}[-1]^2 = V$.

Remark 1 (General I^{2}SB schedules). With a time-varying diffusion $\beta(t)$ and $\sigma^2(t) = \int_0^t \beta$, $\bar{\sigma}^2(t) = \int_t^1 \beta$, the same computation gives $\boldsymbol{x}_t = \mu_0(t)\boldsymbol{x}_0 + \mu_1(t)\boldsymbol{x}_1 + \sigma_{\text{sb}}(t)\boldsymbol{\varepsilon}$ with $\mu_0 = \bar{\sigma}^2/(\sigma^2 + \bar{\sigma}^2)$, $\mu_1 = \sigma^2/(\sigma^2 + \bar{\sigma}^2)$ and $\sigma_{\text{sb}}^2 = \sigma^2\bar{\sigma}^2/(\sigma^2 + \bar{\sigma}^2)$. The Brownian bridge is $\beta \equiv V$, giving $\mu_0 = 1 - t$, $\mu_1 = t$. Everything below holds for a general schedule with $(1 - t, t)$ replaced by (μ_0, μ_1) and $V = \sigma^2(1)$; we keep the Brownian form for readability.

1.2 The reverse posterior

Given $\boldsymbol{x}_0$, the bridge is Markov, and on $[0, t]$ it is a Brownian bridge from $\boldsymbol{x}_0$ to $\boldsymbol{x}_t$. For $s < t$ this gives

$$q(\boldsymbol{x}_s \mid \boldsymbol{x}_t, \boldsymbol{x}_0) = \mathcal{CN}\Big((1 - \tfrac{s}{t})\boldsymbol{x}_0 + \tfrac{s}{t}\,\boldsymbol{x}_t,\ V\,\tfrac{s(t-s)}{t}\,\boldsymbol{I}\Big). \tag{3}$$

I^{2}SB's sampler is (3) with $\boldsymbol{x}_0$ replaced by the network's estimate, iterated from $\boldsymbol{x}_1$ down a fixed ladder of times (`sb.base.reverse_sample`, `posterior="ddpm"`).

1.3 What the network learns

I^{2}SB trains a regressor on pairs: draw $(\boldsymbol{x}_0, \boldsymbol{x}_1)$, $t \sim \mathcal{U}[0, 1]$ and $\boldsymbol{x}_t \sim q(\cdot \mid \boldsymbol{x}_0, \boldsymbol{x}_1)$, and minimise

$$\mathcal{L}(\theta) = \mathbb{E}\,\big\|D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) - \boldsymbol{x}_0\big\|^2. \tag{4}$$

The minimiser of a squared loss is the conditional expectation, so at the optimum

$$D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1]. \tag{5}$$

The conditioning set is whatever the network is given. Our recon regressors receive $\boldsymbol{x}_1$ (and the measurement $\boldsymbol{y}$, $\boldsymbol{E}$) besides $\boldsymbol{x}_t$, so (5) conditions on $\boldsymbol{x}_1$. A network that saw $\boldsymbol{x}_t$ alone would learn $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t]$, which differs whenever $\boldsymbol{x}_1$ carries information about $\boldsymbol{x}_0$ not already in $\boldsymbol{x}_t$.

2 The bridge as a ladder of denoising problems

Lemma 1 (Debiased state)**.** *Fix $\boldsymbol{x}_1$ and $t \in [0, 1)$, and define*

$$\boldsymbol{u}_t \coloneqq \frac{\boldsymbol{x}_t - t\,\boldsymbol{x}_1}{1 - t}. \tag{6}$$

Then

$$\boldsymbol{u}_t = \boldsymbol{x}_0 + \sigma(t)\,\boldsymbol{\varepsilon}, \qquad \sigma(t)^2 = \frac{\sigma_{\text{sb}}(t)^2}{(1-t)^2} = \frac{V\,t}{1-t}, \qquad \boldsymbol{\varepsilon} \perp (\boldsymbol{x}_0, \boldsymbol{x}_1). \tag{7}$$

The map $t \mapsto \sigma(t)$ is a bijection $[0, 1) \to [0, \infty)$ with inverse

$$t(\sigma) = \frac{\sigma^2}{\sigma^2 + V}, \qquad \text{and} \qquad \sigma_{\text{sb}}(t) = (1 - t)\,\sigma(t). \tag{8}$$

Proof. Substitute (2) into (6). Inverting $\sigma^2 = Vt/(1 - t)$ gives $t = \sigma^2/(\sigma^2 + V)$. □

So the bridge state at time t is, given $\boldsymbol{x}_1$, the clean target $\boldsymbol{x}_0$ plus white Gaussian noise of a *known* level $\sigma(t)$. The bridge time and the noise level are one variable. Moving from $t = 1$ to $t = 0$ is the familiar denoising ladder from $\sigma = \infty$ to $\sigma = 0$, with the endpoint $\boldsymbol{x}_1$ acting as side information. Since $(\boldsymbol{x}_t, \boldsymbol{x}_1) \mapsto (\boldsymbol{u}_t, \boldsymbol{x}_1)$ is invertible for $t < 1$,

$$D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1] = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}_t, \boldsymbol{x}_1]: \tag{9}$$

the I^{2}SB regressor is a conditional MMSE denoiser at noise level $\sigma(t)$. $\boldsymbol{x}_1$ enters in two roles: as conditioning information, and in the change of coordinates that removes the deterministic part $t\,\boldsymbol{x}_1$ of the state.

3 MAP estimation given x_1, and Tweedie's formula

3.1 The goal

Given the starting point $\boldsymbol{x}_1$, we want the maximum a posteriori estimate

$$\hat{\boldsymbol{x}}_0 = \arg\max_{\boldsymbol{x}} \ \log p(\boldsymbol{x} \mid \boldsymbol{x}_1), \tag{10}$$

and in reconstruction, with the measurement $\boldsymbol{y}$ too, $\arg\max_{\boldsymbol{x}} \log p(\boldsymbol{x} \mid \boldsymbol{x}_1, \boldsymbol{y})$ (Section 7). The conditional prior $p(\boldsymbol{x} \mid \boldsymbol{x}_1)$ is not available in closed form. What we have is (9), a denoiser of $\boldsymbol{x}_0$ at every noise level, given $\boldsymbol{x}_1$.

3.2 The smoothed conditional prior

Let φ_σ be the $\mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$ density. The law of $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ given $\boldsymbol{x}_1$ is the conditional prior convolved with the noise:

$$p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1) = \int \varphi_\sigma(\boldsymbol{u} - \boldsymbol{x})\, p(\boldsymbol{x} \mid \boldsymbol{x}_1)\, d\boldsymbol{x}, \qquad p_\sigma(\cdot \mid \boldsymbol{x}_1) \xrightarrow[\sigma \to 0]{} p(\cdot \mid \boldsymbol{x}_1). \tag{11}$$

For large σ it is a heavily blurred version of the prior, smooth and close to unimodal. As $\sigma \to 0$ it sharpens back to $p(\cdot \mid \boldsymbol{x}_1)$.

Proposition 1 (Conditional Tweedie / Miyasawa)**.** *For $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I})$ independent of $(\boldsymbol{x}_0, \boldsymbol{x}_1)$,*

$$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] = \boldsymbol{u} + \sigma^2\, \nabla_{\boldsymbol{u}} \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1). \tag{12}$$

Proof. $\varphi_\sigma(\boldsymbol{z}) \propto \exp(-\|\boldsymbol{z}\|^2/\sigma^2)$, so $\nabla_{\boldsymbol{z}} \varphi_\sigma(\boldsymbol{z}) = -\sigma^{-2} \boldsymbol{z}\, \varphi_\sigma(\boldsymbol{z})$ (Wirtinger derivative with respect to $\bar{\boldsymbol{z}}$). Differentiate (11) under the integral:

$$\sigma^2 \nabla_{\boldsymbol{u}} p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1) = \int (\boldsymbol{x} - \boldsymbol{u})\, \varphi_\sigma(\boldsymbol{u} - \boldsymbol{x})\, p(\boldsymbol{x} \mid \boldsymbol{x}_1)\, d\boldsymbol{x} = p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1)\big(\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] - \boldsymbol{u}\big),$$

and divide by $p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1)$. □

Combined with (9), the residual of the trained network *is* the score of the smoothed conditional prior, scaled by the noise variance:

$$\boxed{D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) - \boldsymbol{u}_t = \sigma(t)^2\, \nabla_{\boldsymbol{u}} \log p_{\sigma(t)}(\boldsymbol{u}_t \mid \boldsymbol{x}_1).} \tag{13}$$

A step $\boldsymbol{u} + h\,(D_\theta - \boldsymbol{u})$ is gradient ascent on $\log p_\sigma(\cdot \mid \boldsymbol{x}_1)$ with step size $h\sigma^2$. The step $h = 1$ jumps straight to the posterior mean, which is what a one-shot reconstruction network does.

3.3 Annealing

Ascending $\log p_\sigma$ while lowering σ is a coarse-to-fine continuation. At large σ the landscape is smooth and the ascent is easy. As σ shrinks the iterate follows a mode of the sharpening density, and as $\sigma \to 0$ it approaches a maximiser of $p(\cdot \mid \boldsymbol{x}_1)$, the MAP estimate (10). Injecting fresh noise along the way (Section 4) turns the deterministic ascent into a stochastic one that draws samples instead (Kadkhodaie–Simoncelli [2]). The bridge supplies the whole family $\{p_\sigma(\cdot \mid \boldsymbol{x}_1)\}_\sigma$ through a single network, one noise level per bridge time.

4 The Tweedie (Kadkhodaie–Simoncelli / ImMAP) ascent

In debiased coordinates the iterate is a pair $(\boldsymbol{u}_k, \sigma_k)$. One step reads

$$\boldsymbol{v}_k = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}_k, \boldsymbol{x}_1] \approx D_\theta(\boldsymbol{x}_k, t_k, \boldsymbol{x}_1), \tag{14}$$

$$\boldsymbol{u}_{k+1} = (1-h_k)\,\boldsymbol{u}_k + h_k\,\boldsymbol{v}_k + \gamma_k^u\,\boldsymbol{z}_k \;=\; \boldsymbol{u}_k + h_k\,\sigma_k^2\,\nabla_{\boldsymbol{u}} \log p_{\sigma_k}(\boldsymbol{u}_k \mid \boldsymbol{x}_1) + \gamma_k^u\,\boldsymbol{z}_k, \tag{15}$$

with $\boldsymbol{z}_k \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I})$ fresh and $h_k \in (0, 1]$.

Noise accounting. Suppose $\boldsymbol{v}_k$ removes the noise of $\boldsymbol{u}_k$, i.e. $\boldsymbol{v}_k \approx \boldsymbol{x}_0$. Then the noise left in $(1-h_k)\boldsymbol{u}_k + h_k\boldsymbol{v}_k$ is $(1-h_k)\sigma_k\boldsymbol{\varepsilon}$, and after the injection

$$\sigma_{k+1}^2 = (1-h_k)^2\sigma_k^2 + (\gamma_k^u)^2. \tag{16}$$

We prescribe the contraction $\sigma_{k+1} = (1-\beta h_k)\,\sigma_k$, with $\beta \in (0, 1]$, which fixes the injected noise:

$$(\gamma_k^u)^2 = \left[(1-\beta h_k)^2 - (1-h_k)^2\right]\sigma_k^2. \tag{17}$$

$\beta = 1$ injects nothing and gives the deterministic ascent toward the MAP estimate; $\beta < 1$ keeps fresh noise in every step. β sets how much of the noise removed by a step is put back, and h_k how far the step moves.

Schedules. The step size follows Kadkhodaie–Simoncelli, $h_k = h_0 k/(1 + h_0(k-1))$, small at first and growing toward 1. The noise level either follows the recursion alone ("open"), or is re-estimated from the residual. Since $\mathbb{E}\|D_\theta - \boldsymbol{u}\|^2 \approx N\sigma^2$ when the denoiser is accurate,

$$\hat{\sigma}_k^2 = \tfrac{1}{N}\|\boldsymbol{v}_k - \boldsymbol{u}_k\|^2,$$

which is ImMAP's self-paced ladder. This estimate is biased high at large σ, where D_θ shrinks toward the mean, and includes D_θ's own error at small σ. The iteration stops when $\sigma_k < \sigma_{\min}$ and returns the last $\boldsymbol{v}_k$.

5 Re-placing the step on the bridge

Forming $\boldsymbol{u}$ explicitly is a bad idea near $t = 1$: the factor $1/(1-t)$ in (6) blows up. Instead we carry the step out in the bridge coordinates themselves. After the step the noise level is $\sigma' = (1-\beta h_k)\sigma_k$, i.e. the new bridge time is $t' = \sigma'^2/(\sigma'^2 + V)$ by (8). We map the stepped $\bar{\boldsymbol{u}} = (1-h_k)\boldsymbol{u}_k + h_k\boldsymbol{v}_k$ back onto the bridge *at* t':

$$\boldsymbol{x}_{k+1} = (1-t')\,\bar{\boldsymbol{u}} + t'\,\boldsymbol{x}_1 + \gamma_k\,\boldsymbol{\varepsilon}_k. \tag{18}$$

Proposition 2 (The step in bridge coordinates)**.** *With $\boldsymbol{u}_k = (\boldsymbol{x}_k - t_k\boldsymbol{x}_1)/(1-t_k)$, (18) is*

$$\boldsymbol{x}_{k+1} = c_x\,\boldsymbol{x}_k + c_v\,\boldsymbol{v}_k + c_1\,\boldsymbol{x}_1 + \gamma_k\,\boldsymbol{\varepsilon}_k, \qquad c_x = \frac{(1-h_k)(1-t')}{1-t_k}, \quad c_v = h_k(1-t'), \quad c_1 = t' - c_x\,t_k, \tag{19}$$

with $c_x + c_v + c_1 = 1$. The noise carried over from $\boldsymbol{x}_k$ has std $\rho_k\,\sigma_{\mathrm{sb}}(t')$ with

$$\rho_k = \frac{1-h_k}{1-\beta h_k} \in [0, 1], \tag{20}$$

so the injection that brings the total exactly to the bridge std at t' is

$$\boxed{\gamma_k^2 = \sigma_{\mathrm{sb}}(t')^2\,(1-\rho_k^2) = (1-t')^2(\gamma_k^u)^2.} \tag{21}$$

Proof. Expand (18): $(1-t')\big[(1-h_k)\frac{\boldsymbol{x}_k - t_k\boldsymbol{x}_1}{1-t_k} + h_k\boldsymbol{v}_k\big] + t'\boldsymbol{x}_1$ gives the three coefficients. They sum to $c_x(1-t_k) + h_k(1-t') + t' = (1-h_k)(1-t') + h_k(1-t') + t' = 1$. The noise in $\boldsymbol{x}_k$ is $\sigma_{\rm sb}(t_k)\boldsymbol{\varepsilon} = (1-t_k)\sigma_k\boldsymbol{\varepsilon}$, and c_x times it is $(1-h_k)(1-t')\sigma_k\boldsymbol{\varepsilon} = \rho_k(1-t')\sigma'\boldsymbol{\varepsilon} = \rho_k\,\sigma_{\rm sb}(t')\boldsymbol{\varepsilon}$, using $\sigma' = (1-\beta h_k)\sigma_k$ and $\sigma_{\rm sb}(t') = (1-t')\sigma'$. Adding independent noise of variance γ_k^2 gives total variance $\sigma_{\rm sb}(t')^2$ exactly when (21) holds. The last equality is (17) times $(1-t')^2$. □

Remark 2 (The iterates stay on the bridge). If $\boldsymbol{v}_k = \boldsymbol{x}_0$ and $\boldsymbol{x}_k \sim q(\cdot \mid \boldsymbol{x}_0, \boldsymbol{x}_1)$ at t_k, then $\boldsymbol{x}_{k+1} = (1-t')\boldsymbol{x}_0 + t'\boldsymbol{x}_1 + \rho_k\sigma_{\rm sb}(t')\boldsymbol{\varepsilon} + \gamma_k\boldsymbol{\varepsilon}_k$ is distributed exactly as the bridge marginal (1) at t'. With an accurate denoiser, every iterate is a bridge state, which is the input distribution the network was trained on.

Remark 3 (Why the step must land at the *new* time). An earlier variant relaxed $\boldsymbol{x}_k$ toward the bridge mean at the *current* time, $(1-t_k)\boldsymbol{v}_k + t_k\boldsymbol{x}_1$. On a noise-free iterate that already lies on the bridge, with a perfect denoiser, that mean equals $\boldsymbol{x}_k$ and the time re-estimated from it returns t_k again. The step then removes noise but never moves along the bridge, and the iteration stalls. Mapping onto the bridge at the contracted time t' is what makes progress.

Remark 4 (Starting point). $\boldsymbol{x}_1$ itself is a fixed point: at $t = 1$ the state carries no noise and $\sigma = \infty$, so any bridge-consistent estimator maps it to itself. The ascent therefore starts at a finite σ_0, i.e. $t_0 = \sigma_0^2/(\sigma_0^2 + V) < 1$, as Kadkhodaie–Simoncelli start at a finite σ_0. A natural initial state is the bridge sample around a first estimate, $\boldsymbol{x}_{t_0} = (1-t_0)\,D_\theta(\boldsymbol{x}_1, 1, \boldsymbol{x}_1) + t_0\boldsymbol{x}_1 + \sigma_{\rm sb}(t_0)\boldsymbol{\varepsilon}$.

Algorithm 1 ImMAP-SB: Tweedie ascent on the bridge (`sb/immap_sb_ascent.py`)

Require: endpoint $\boldsymbol{x}_1$; regressor D_θ; total variance $V = 4\tau^2$; start σ_0; $\beta \in (0, 1]$; h_0; $\sigma_{\min}$; optional data prox P (Sec. 7)

$t_0 \leftarrow \sigma_0^2/(\sigma_0^2 + V)$; $\quad \boldsymbol{x} \leftarrow (1-t_0)\,D_\theta(\boldsymbol{x}_1, 1, \boldsymbol{x}_1) + t_0\boldsymbol{x}_1 + \sigma_{\rm sb}(t_0)\,\boldsymbol{\varepsilon}$; $\quad k \leftarrow 1$

while $\sigma_k > \sigma_{\min}$ **do**

  $\boldsymbol{v} \leftarrow D_\theta(\boldsymbol{x}, t_k, \boldsymbol{x}_1)$; $\quad$ if P: $\boldsymbol{v} \leftarrow P(\boldsymbol{v};\, \sigma_k)$ $\quad\triangleright$ conditional MMSE estimate

  $h \leftarrow h_0 k/(1 + h_0(k-1))$; $\quad \sigma' \leftarrow (1-\beta h)\,\sigma_k$; $\quad t' \leftarrow \sigma'^2/(\sigma'^2 + V)$

  $c_x \leftarrow (1-h)(1-t')/(1-t_k)$; $\quad c_v \leftarrow h(1-t')$; $\quad c_1 \leftarrow t' - c_x t_k$

  $\rho \leftarrow (1-h)/(1-\beta h)$; $\quad \gamma \leftarrow \sigma_{\rm sb}(t')\sqrt{1-\rho^2}$

  $\boldsymbol{x} \leftarrow c_x\,\boldsymbol{x} + c_v\,\boldsymbol{v} + c_1\,\boldsymbol{x}_1 + \gamma\,\boldsymbol{\varepsilon}$, $\quad \boldsymbol{\varepsilon} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I})$

  $(t_k, \sigma_k) \leftarrow (t', \sigma')$; $\quad k \leftarrow k + 1$ $\quad\triangleright$ or re-estimate σ_k from $\|\boldsymbol{v} - \boldsymbol{u}\|$

end while

return $\boldsymbol{v}$

6 I^{2}SB's sampler is a special case

Proposition 3. *Take one step of the ascent from t to a prescribed $s < t$, and let $r = \sigma(s)/\sigma(t)$. With*

$$h = 1 - r^2 = \frac{t-s}{t(1-s)}, \qquad \beta = \frac{1}{1+r}, \tag{22}$$

Proposition 2 gives exactly I^{2}SB's reverse step (3) with $\boldsymbol{x}_0$ replaced by $\boldsymbol{v} = D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1)$.

Proof. First the time. $1 - \beta h = 1 - (1-r)(1+r)/(1+r) = r$, so $\sigma' = r\,\sigma(t) = \sigma(s)$ and $t' = s$. With $\sigma^2 = Vt/(1-t)$, $r^2 = s(1-t)/(t(1-s))$, which gives the stated form of h. Then the coefficients of (19):

$$c_v = h(1-s) = \frac{t-s}{t} = 1 - \frac{s}{t}, \qquad c_x = \frac{r^2(1-s)}{1-t} = \frac{s}{t}, \qquad c_1 = s - \frac{s}{t}\,t = 0,$$

the mean of (3). Finally the noise. $\rho = (1-h)/(1-\beta h) = r^2/r = r$, so

$$\gamma^2 = \sigma_{\mathrm{sb}}(s)^2(1-r^2) = V s(1-s)\cdot\frac{t-s}{t(1-s)} = V\,\frac{s(t-s)}{t},$$

the variance of (3). □

So I^{2}SB's sampler is the Tweedie ascent with its step size tied to the training ladder, $h = 1-r^2$, and with $\beta = 1/(1+r)$. For small steps, $r \to 1$ and $\beta \to \frac{1}{2}$: the I^{2}SB posterior re-injects roughly half the noise each step removes. This is a stochastic, sampling-type ascent, not the deterministic one ($\beta = 1$). The ImMAP view separates three things the I^{2}SB sampler ties together:

- the **step size** h_k, which can be chosen freely instead of being set by the ladder;
- the **stochasticity** β, anywhere from deterministic MAP ascent ($\beta = 1$) to I^{2}SB's $\approx \frac{1}{2}$;
- the **noise ladder**: open-loop, or self-paced from the residual norm, with a starting level σ_0 that need not be the trained maximum.

Proposition 3 is checked numerically against `sb.base.reverse_sample` in `tests/test_immap_sb_ascent.py`.

7 Adding the measurement: a data-consistency prox

For reconstruction the target is $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1, \boldsymbol{y})$, with $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x}_0 + \boldsymbol{n}$ and $\boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \sigma_y^2\boldsymbol{I})$. Take the local Gaussian approximation $p(\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1) \approx \mathcal{CN}(\boldsymbol{v}, c\,\sigma^2\boldsymbol{I})$ around the denoiser output. Here $c \le 1$ allows for the posterior spread being smaller than the noise variance, and is hand-tuned as ImMAP's λ. Then

$$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}] \approx \arg\min_{\boldsymbol{x}} \frac{\|\boldsymbol{y}-\boldsymbol{E}\boldsymbol{x}\|^2}{\sigma_y^2} + \frac{\|\boldsymbol{x}-\boldsymbol{v}\|^2}{c\,\sigma^2} = (\boldsymbol{E}^H\boldsymbol{E} + \lambda\boldsymbol{I})^{-1}(\boldsymbol{E}^H\boldsymbol{y} + \lambda\,\boldsymbol{v}), \qquad \lambda = \frac{\sigma_y^2}{c\,\sigma^2}. \tag{23}$$

It is solved with a few conjugate-gradient iterations warm-started at $\boldsymbol{v}$, and replaces $\boldsymbol{v}_k$ in Algorithm 1.

- λ **follows the bridge.** It is large at small σ, where the denoiser is trusted and the prox is nearly the identity, and small at large σ, where the data dominate. The CG step of I^{2}SB-Recon [5] is (23) with a *fixed* λ ($7\cdot 10^{-5}$). With measurement noise $\sigma_y > 0$, a fixed tiny λ fits the noise on the measured lines at every step and amplifies it. The ImMAP view sets λ from σ_y and $\sigma(t)$ instead.
- **No double counting in principle.** When $\boldsymbol{x}_1 = \boldsymbol{E}^H\boldsymbol{y}$ and the noise is white, $\boldsymbol{E}^H\boldsymbol{y}$ is a sufficient statistic for $\boldsymbol{x}_0$ given $\boldsymbol{y}$, so a regressor conditioned on $\boldsymbol{x}_1$ already targets $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{y}]$ and (23) adds no new information. In practice it corrects the regressor's errors, which is where its value lies. When $\boldsymbol{x}_1$ does *not* contain the measurement (a guide-contrast start, or another study), (23) is the only place $\boldsymbol{y}$ enters.

8 Summary

The reading in one line: *with $\boldsymbol{x}_1$ known, the bridge is a family of conditional denoising problems; the I^{2}SB network is their MMSE denoiser; Tweedie turns it into the score of the smoothed conditional prior; and annealed ascent on that score, re-placed on the bridge at each new time, approaches the MAP estimate of $\boldsymbol{x}_0$ given $\boldsymbol{x}_1$, with I^{2}SB's own sampler as one setting of its step size and noise.*

Object	Formula	Code
bridge marginal	$\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1 + \sigma_{\text{sb}}(t)\boldsymbol{\varepsilon}$, $\sigma_{\text{sb}}^2 = Vt(1-t)$	`sb.base.bridge_coeffs`
temperature	$V = 4\tau^2$, $\max_t \sigma_{\text{sb}} = \tau$	`std_fwd[-1]**2`
debiased state	$\boldsymbol{u}_t = (\boldsymbol{x}_t - t\boldsymbol{x}_1)/(1-t) = \boldsymbol{x}_0 + \sigma(t)\boldsymbol{\varepsilon}$	`_Bridge`
time $\leftrightarrow$ noise	$\sigma^2 = Vt/(1-t)$, $t = \sigma^2/(\sigma^2+V)$	`t_of_sigma`, `sigma_of_t`
regressor	$D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}_t, \boldsymbol{x}_1]$	`predict_x0_recon`
Tweedie	$D_\theta - \boldsymbol{u} = \sigma^2 \nabla \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1)$	—
ascent step	$\boldsymbol{x}' = c_x\boldsymbol{x} + c_v\boldsymbol{v} + c_1\boldsymbol{x}_1 + \gamma\boldsymbol{\varepsilon}$	`bridge_ascent_step`
injection	$\gamma^2 = \sigma_{\text{sb}}(t')^2(1-\rho^2)$, $\rho = \frac{1-h}{1-\beta h}$	`bridge_ascent_step`
I^{2}SB special case	$h = 1 - r^2$, $\beta = 1/(1+r)$	`reverse_sample("ddpm")`
data prox	$(\boldsymbol{E}^H\boldsymbol{E} + \lambda\boldsymbol{I})^{-1}(\boldsymbol{E}^H\boldsymbol{y} + \lambda\boldsymbol{v})$, $\lambda = \sigma_y^2/(c\sigma^2)$	`cg_data_consistency`, `ImMAPProx`

Table 1: The ImMAP reading of I^{2}SB and where each piece lives in the code (`sb/base.py`, `sb/immap_sb_ascent.py`, `sb/i2sb_recon.py`, `sb/immap_sb.py`).

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